BUILT · WORKING

Running it safely `runner/`

Code written by an AI has to actually run before anyone can believe it. This stage executes it in a sealed box and reports what happened.

WHAT GOES IN, WHAT COMES OUT

INPUT	WHAT HAPPENS	OUTPUT
The generated script from stage 3	**Runs inside a sealed container** cheap checks first, then longer ones	**Scores, logs, a verdict** and, on failure, a diagnosis

THE KEY POINTS

- **Nothing runs on the real computer.** The script is executed inside a sealed container — code written by an AI, running unsupervised, should not have access to anything it can damage.
- **It starts cheap and escalates.** First a few seconds to check the thing runs at all, then one short pass, then a longer one. A broken script fails in seconds instead of hours.
- **It knows nothing about the script it runs.** It only ever says “run this, at this size”. That means a completely different paper needs no changes here at all.
- **When something fails, one cheap call sorts out whose fault it is** — a bug in the generated code, or a problem with the container. The two need opposite responses, so the distinction matters.
- **When it succeeds, nothing is asked.** A successful run needs no explanation, so no extra call is made — the common path costs nothing.

WHAT COMES BACK

A REAL RUN — NETWORK IN NETWORK

```
[probe] PASSED in 32.2s
[smoke] PASSED in 62.1s
        Test Error = 86.7%
[loop]  VERDICT: success
```

That 86.7% is not a failure. The paper claims 10.41%, but this run trained for one pass over a few hundred images — a few seconds of work. It answers “does this run at all”, never “is the number right”. Getting a real number needs a real training run, which is the problem below.

MEASURED

2
papers run successfully

1.7 GB
the container image

22 days
for one real training run

The blocker, in one number. On this laptop's processor, training one of these models properly — the full run the paper actually did — would take about **22 days**. For one paper, for one claimed result.

A graphics card would cut that to hours. Until then, every run is a short check that the machinery works, not a real reproduction.